

Table 3: **In-context alignment with *multi-turn* examples.** Second-turn answers for all methods are generated using GPT-4o. Extending URIAL with two-turn examples improves the 2nd-turn MT-Bench score. This shows that in-context alignment is not limited to single-turn performance.

In-context prompt	Mistral-7B-v0.2			Llama-3.1-8B		
	Turn 1	Turn 2	Avg	Turn 1	Turn 2	Avg
Base model (no prompt)	6.52	5.68	6.10	6.25	5.17	5.71
Original URIAL (single-turn)	6.99	5.55	6.27	6.95	5.31	6.13
Two-turn URIAL	7.18	5.68	6.43	7.31	6.44	6.87
URIAL + 3 two-turn SkillMix examples (no separation tag)	7.46	5.88	6.67	7.21	6.43	6.82
URIAL + 3 two-turn SkillMix examples (separation tag)	7.20	6.31	6.76	7.22	6.71	6.97
URIAL + 6 two-turn SkillMix examples (no separation tag)	7.18	5.89	6.53	7.14	6.26	6.70
URIAL + 6 two-turn SkillMix examples (separation tag)	7.24	5.78	6.51	6.89	6.33	6.61

the base LLMs can handle multi-turn conversations even with only having single-turn demonstrations in the context, but it does not quantify the performance nor rigorously study the role of examples that have more than one turn of conversation. To bridge this research gap, we further test how the presence of multi-turn examples in the context influences the instruction-following performance of in-context aligned models.

Experimental setup. We experiment using two open source LLMs, Mistral-7B-v0.2 and Llama-3.1-8B, and evaluate on MT-Bench. To ensure the multi-turn examples we test have high quality, we select single-turn examples randomly from SkillMix as the basis, the first turn, and then prompt GPT-4o to ask a follow-up question after seeing the first-turn conversation and generate an answer by using an instruction "Here is a single-turn conversation between a user and an assistant, please ask a follow-up question and provide an answer.". We examine the influence of two-turn examples from two perspectives, the number of two-turn examples (3 vs 6), and the format, with or without separation tags like "Here are 3 single-turn/multi-turn examples:."

Results. In Table 3, we observe that extending 3 URIAL single-turn examples to 2-turn examples consistently improves the evaluation scores on both turns, showing that in-context alignment solely with multi-turn examples can take care of single-turn conversations. The presence of two-turn examples in the context always leads to improved 2nd-turn MT-Bench scores, for example, an increase from 5.31 to 6.43 after adding 3 two-turn SkillMix examples on top of URIAL on Llama-3.1-8B. Another finding from our experiments is that the format of placing demonstrations plays a pivotal role in multi-turn interactions. In particular, adding a separation tag to arrange demonstrations in two groups improves 2nd-turn scores, from 5.88 to 6.31 on Mistral-7B-v0.2 and from 6.43 to 6.71 on Llama-3.1-8B, although sometimes it hurts the 1st-turn performance. However, we find that increasing the number of two-turn examples in the context does not lead to more benefits with or without a better format.

3 MANY-SHOT IN-CONTEXT LEARNING FOR INSTRUCTION FOLLOWING

Sec. 2.1 indicate that URIAL alone is not able, in most cases, to reach the performance of instruct models. In the following, we explore whether we can close this gap. We focus on Mistral-7B-v0.2 (Jiang et al., 2023) and Llama-3.1-8B (Dubey et al., 2024) since (a) both URIAL and the aligned model achieve competitive performance in Table 1, and (b) these base models have context windows of 32k and 128k respectively, which can fit many examples (around 50 for Mistral-7B-v0.2 and more than 200 for Llama-3.1-8B). Details on the experimental setup are in App. A.

Data. We adopt the SkillMix (Kaur et al., 2024) dataset consisting 4,000 examples as the source of high-quality examples for our many-shot in-context alignment experiments. The query-answer paired data generation process of SkillMix explicitly combines diversity and difficulty through random skill combinations, where the set of core skills for instruction-following, such as developing training programs and digital proficiency, are extracted by prompting a powerful LLM like GPT-4-Turbo. We present more details of SkillMix, including some examples (Fig. 6 and Fig. 7), in App. A.1.

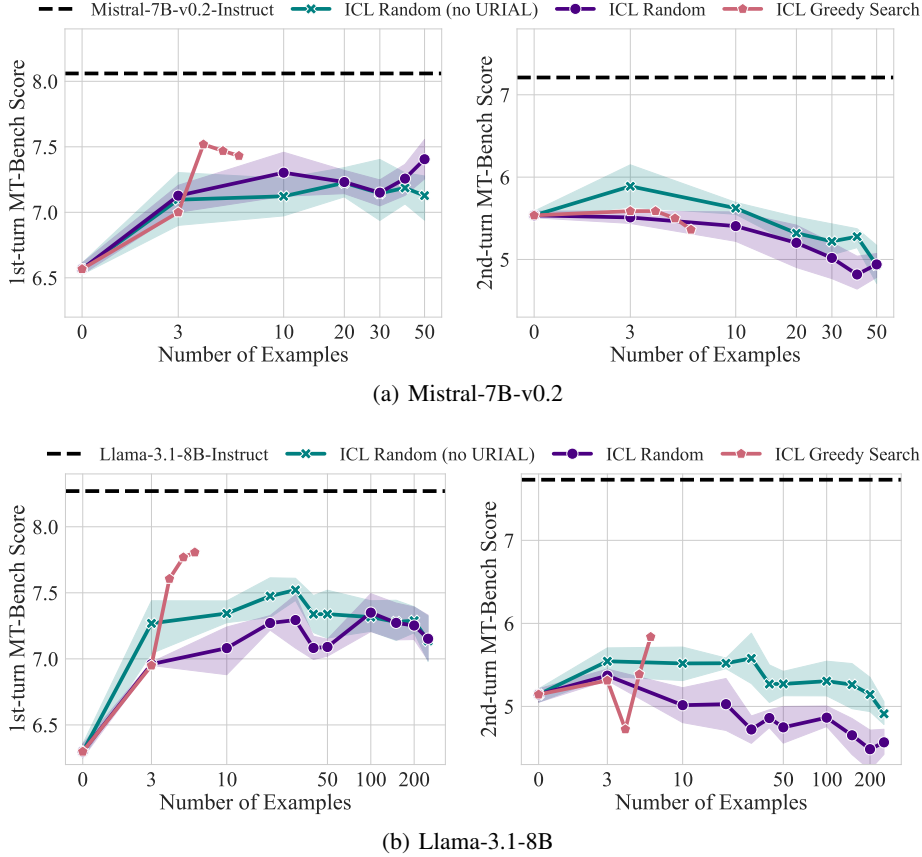


Figure 3: **Scaling the number of demonstrations for alignment with ICL on Mistral-7B-v0.2 and Llama-3.1-8B.** We measure the alignment performance of different settings using the MT-Bench score. ICL with more demonstrations quickly saturates and does not bridge the performance gap between the base model and its aligned counterpart. In particular, the ICL alignment performance of 3 random examples from the high-quality SkillMix dataset surpasses that of 3 examples from URIAL.

3.1 SCALING UP THE NUMBER OF IN-CONTEXT DEMONSTRATIONS

Given the success of many-shot ICL (Agarwal et al., 2024), we test the effect of increasing the number of in-context demonstrations. We sample random examples from the SkillMix dataset, since it contains high-quality question-answer pairs. We consider two scenarios: (1) all demonstrations are randomly chosen from SkillMix (the set of rules from URIAL is kept), and (2) we add the new demonstrations on top of the URIAL examples. In Fig. 3, we report the results on MT-Bench when varying the number of demonstrations. We repeat sampling with 5 random seeds and show mean and standard deviation over the corresponding results. We observe that for both base LLMs, the 1st-turn score increases with 3, 10 and 20 examples, but then plateaus without clear benefits from scaling up beyond 30 demonstrations (for Llama-3.1-8B the trend becomes even slightly negative with more than 100 examples). Thus, we find that adding up to 30 high-quality examples from SkillMix can improve alignment via ICL. This result contrasts with the findings of Lin et al. (2024), which showed that URIAL performed better with 3 demonstrations than with 8.

Next, we observe that in-context alignment does not improve the second-turn score. In fact, adding more examples continues to decrease performance, even falling below that of the base models. This behavior is likely due to the presence of only single-turn examples in the prompt, causing the LLM to respond in the same style without adapting to more complex conversations. Overall, these results show that, unlike in the setting of Agarwal et al. (2024); Bertsch et al. (2024), simply scaling the number of ICL examples is not sufficient to consistently improve the instruction following performance.

Finally, we notice that on Llama-3.1-8B, using three examples from SkillMix attain, on average, better performance than using the three examples of URIAL. Overall, there is no significant difference

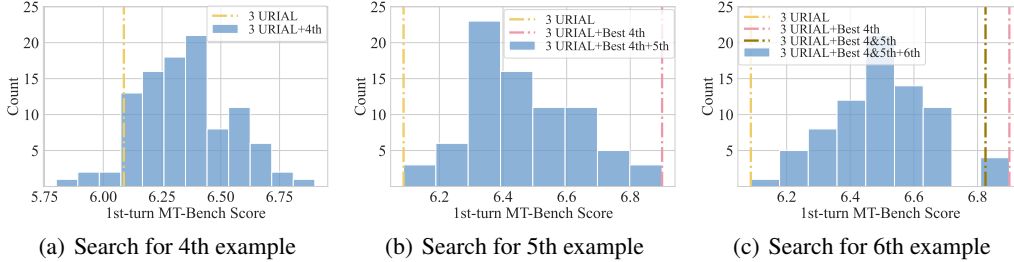


Figure 4: **The distribution of the 1st-turn MT-Bench score (GPT-4-Turbo as judge) on Llama-3.1-8B obtained by adding multiple instructions from SkillMix (Kaur et al., 2024) as a 4th (a), 5th (b), 6th (c) demonstration to URIAL.** The dashed lines of various colors refer to the 1st-turn MT-Bench score of the obtained searching results. A majority of the 4th examples contribute positively to the model’s instruction-following performance, but the improvement quickly diminishes when running the greedy search for 5th and 6th demonstrations.

Table 4: **Improved in-context (IC) alignment by adding to URIAL prompt the demonstrations found via greedy search.** We find the optimal IC prompts in an incremental way by selecting query-response pairs from the high-quality SkillMix dataset. We report the 1st-turn score on MT-Bench and the length-controlled (LC) win rates on AlpacaEval 2.0 when using different IC prompts.

Model	Mistral-7B-v0.2		Llama-3.1-8B	
	MT-Bench (1st)	AlpacaEval 2.0	MT-Bench (1st)	AlpacaEval 2.0
URIAL (3 examples)	7.00	8.22	6.95	7.28
URIAL + greedy search (1 ex.)	7.52	7.53	7.61	8.61
URIAL + greedy search (2 ex.)	7.47	7.78	7.77	8.16
URIAL + greedy search (3 ex.)	7.43	8.55	7.81	8.19

in the scaling behavior between using or not using the URIAL demonstrations. Thus, we conclude that the success of in-context alignment is only partially dependent on the demonstrations themselves, provided they are of sufficient quality (see also Sec. 4). We also include the results of scaling experiments (Sec. E.1) conducted on AlpacaEval 2.0, which consists of 805 examples.

3.2 GREEDY SEARCH FOR EFFECTIVE DEMONSTRATIONS

Given the limited success of adding random demonstrations to URIAL, we try to greedily maximize the MT-Bench score by testing 100 high quality instructions from SkillMix as the 4th additional example to URIAL. For each resulting ICL prompt, we compute the MT-Bench score with GPT-4-Turbo as judge instead of GPT-4 (used in MT-Bench) as the former is faster, cheaper, and helps mitigate potential overfitting to the benchmark score. We then repeat this procedure sequentially to find a 5th and a 6th demonstration, restricting the search space at each round to only instructions leading to a high enough MT-Bench score to reduce the computational cost (see details in App. A.3).

We add the results of the greedy search to the plots in Fig. 3, computing the true MT-Bench (i.e., with GPT-4 as judge). For both base models, the 4th example found by greedy search is sufficient to match the best 1st-turn score achieved by scaling the in-context examples with random demonstrations (see Sec. 3.1). For Llama-3.1-8B, the 5th and 6th demonstrations further improve the score, while they are not helpful for Mistral-7B-v0.2. In Table 4, we provide the details of these results: the 4th ICL example yields a significant improvement over URIAL, e.g., from 6.95 to 7.61 for Llama-3.1-8B, with only a further +0.20 given by the 5th and 6th. Overall, this evaluation further indicates that the improvement in instruction following one can achieve with in-context alignment quickly saturates when increasing the number of examples. In Fig. 4, we show the distribution of the scores (with GPT-4-Turbo as judge) at step of the greedy search when using Llama-3.1-8B as base model (the analog for Mistral-7B-v0.2 in App. E.2). Most demonstrations improve the score when added as the 4th example on top of URIAL (as shown in the first plot), but no further progress is observed with additional demonstrations.

We further test the IC prompts found by the greedy search on the AlpacaEval 2.0 (Li et al., 2023a) benchmark, where we measure length-controlled win rate. As shown in Table 4, using the examples

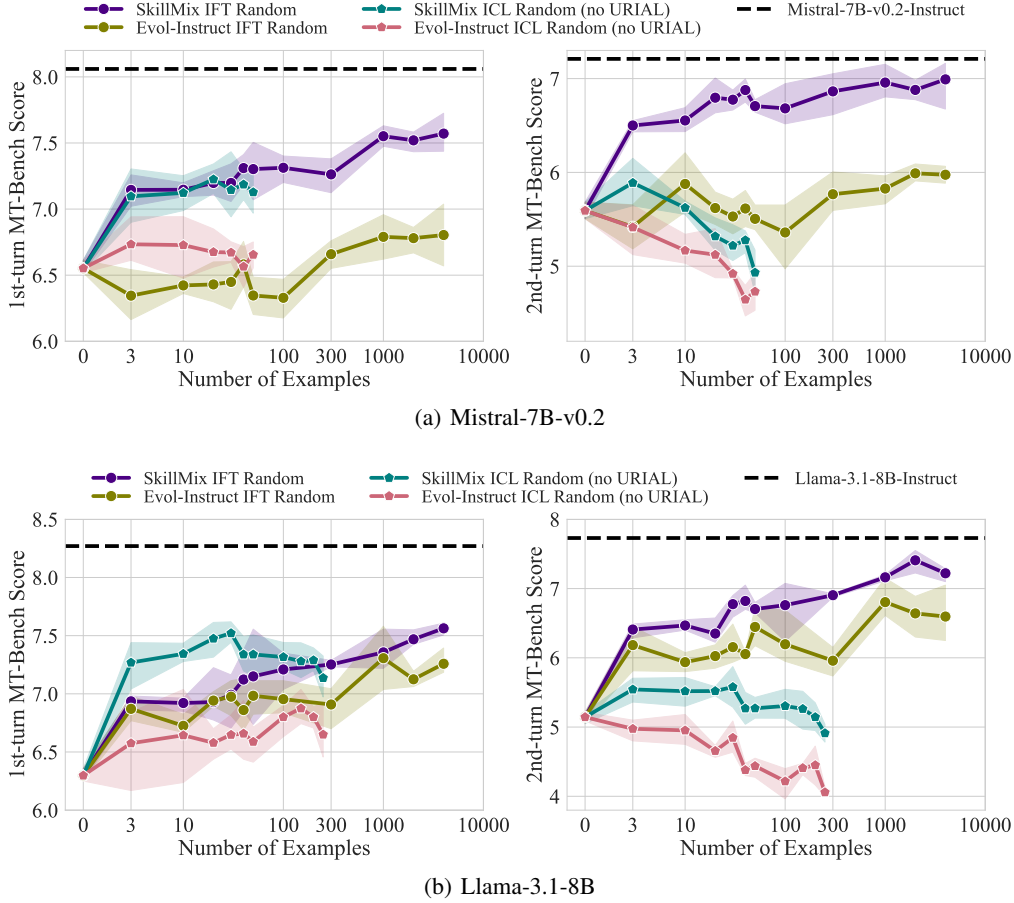


Figure 5: **Comparison of ICL vs IFT for alignment in the low data regime.** We measure the alignment performance of different settings for Mistral-7B-v0.2 and Llama-3.1-8B using the MT-Bench score. IFT with more demonstrations keeps improving the alignment performance, almost bridging the gap between the base model and its aligned counterpart. IFT-aligned models perform well on multi-turn conversations, unlike with ICL. Finally, data quality has significant impact on both IFT and ICL: the higher-quality SkillMix leads to better performance than Evol-Instruct.

given by greedy search leads to better results than plain URIAL on AlpacaEval 2.0, without directly optimizing for it. This improvement indicates that the found IC prompt does not completely overfit to the MT-Bench score, although the gains are less consistent than on MT-Bench. For Mistral-7B-v0.2, URIAL with 3 greedy search examples is the only configuration that outperforms URIAL, while for Llama-3.1-8B just one additional demonstration yields the best win rate, with a significant improvement over URIAL (7.28 vs 8.61).

4 A COMPARISON OF ICL VS IFT FOR INSTRUCTION FOLLOWING

The results from the previous sections strongly suggest that in-context alignment might not be as competitive as sophisticated, computationally heavy alignment techniques. However, it is unclear how it compares to more lightweight approaches, such as IFT, when applied to a small number of examples. In the low-data regime, the choice between ICL and IFT can be viewed as a decision between allocating resources for fine-tuning (which permanently modifies the model’s weights) or increasing inference time by adding the in-context prompt each time.

Setup. We test IFT on the same number of instructions from SkillMix as used in the many-shot ICL experiments in Sec. 3, ranging from 3 to 250 (the maximum allowed by the context length of the base models). Moreover, we scale the dataset up to 4k, that is the entire SkillMix. Each training run is repeated over multiple seeds, resulting in different training datasets (more details in App. A.4).

Results on Mistral-7B-v0.2. The top part of Fig. 5 shows that IFT and ICL on SkillMix (violet and green curves respectively) perform almost identically in 1st-turn score until 50 examples are used. Beyond this point, increasing the number of training examples consistently improves the performance of IFT. The strong performance of IFT in the low-data regime is particularly surprising, as one might expect overfitting when training on very few examples (as few as 3) over several epochs, yet this does not occur. Finally, ICL and IFT show nearly opposite trends for the 2nd-turn score: increasing the dataset size benefits IFT (almost reaching the performance of the instruct model) but detrimental to ICL. This behavior suggests that, while IFT is less flexible due to the model weights being updated, it generalizes better to tasks different from those used for alignment (recall that the instructions in SkillMix are single-turn only).

Results on Llama-3.1-7B. As shown in the bottom part of Fig. 5, ICL outperforms IFT when using between 3 and 30 demonstrations, though IFT remains effective even with these few examples on this base model. However, we note that IFT matches or exceeds the best performance of ICL (obtained with 20-30 examples) only when using 2k or 4k examples (the entire SkillMix), which represents two orders of magnitude more data. This result confirms that ICL with high-quality data is a viable alternative to IFT when only a limited number of demonstrations are available. Finally, the observations for the 2nd-turn scores are consistent with those for Mistral-7B-v0.2, with IFT consistently outperforming ICL.

Effect of data quality. Finally, we test the effect of using lower-quality instructions compared to SkillMix. In this experiment, we repeat the ICL vs. IFT comparison using random demonstrations from Evol-Instruct-70k (Xu et al., 2024), and show the results in Fig. 5. With Evol-Instruct data, both ICL (red curve) and IFT (yellow curve) perform significantly worse than their counterparts using SkillMix. This trend is consistent across the number of examples, base models, and for both single-turn and multi-turn instructions. Interestingly, the performance gap between IFT on SkillMix and Evol-Instruct is significantly smaller on Llama-3.1-8B than on Mistral-7B-v0.2, suggesting that better pre-trained models may partially compensate for lower-quality fine-tuning data. Finally, we notice that ICL with 3 examples from Evol-Instruct gets worse 1st-turn scores than URIAL (with also 3 examples), whereas ICL with 3 examples from SkillMix matches (on Mistral-7B-v0.2) or outperforms (on Llama-3.1-8B) URIAL (see also discussion in Sec. 3.1). These observations further confirm the importance of instruction quality for alignment, whether in the context of in-context learning or instruction fine-tuning.

5 CONCLUSIONS

In this work, we have first illustrated that ICL via URIAL is a good baseline for instruction-following alignment, but with a few limitations: it typically performs slightly worse than IFT on single-turn conversations and does not generalize well to multi-round ones. Then, we have uncovered the key components for IC alignment: e.g. the decoding parameters alone can, surprisingly, make base models reasonably good at instruction following. Adding in-context high-quality demonstrations improves performance beyond what previously suggested (Lin et al., 2024), but not enough to reach LLMs aligned with sophisticated methods (e.g., RLHF). We conjecture that via ICL, the LLM can learn to infer the response style, but the overall learning signal is not sufficiently strong to benefit from a large amount of examples, despite the long context windows of recent LLMs.

Moreover, to the best of our knowledge, we have provided the first systematic comparison of ICL and IFT for instruction following when using the same (small) number of demonstrations. Surprisingly, the two approaches are roughly equivalent in terms of single-turn conversation performance. However, models trained via IFT, unlike ICL, can generalize to multi-round conversations, which are out-of-distribution compared to the examples used for alignment. Overall, our work provides a deeper and more complete understanding of how in-context alignment works, as well as of its potential and limitations, in particular in comparison to fine-tuning. This comprehension might be the basis for future work aimed at using ICL to efficiently customize LLMs without the need of fine-tuning, as well as exploring the fundamental differences between base and aligned models.

Limitations. We perform greedy search with limited computational resources, e.g. the set of candidate examples is relatively small. We expect that more effective ICL prompts could be identified with additional resources. Moreover, our focus is on instruction-following tasks, but other types of alignment, such as safety-oriented tasks, may also be of interest.